Research project plan

Hub and accountable project authority

synthetic-team-demo

Factory 2.6.0 | 2026-10-25T01:43:11+00:00

DRAFT FOR REVIEW - NOT PREPARATION APPROVAL

40 of 43 responses recorded | 3 unresolved | 4 kits

Configuration SHA-256: 8bd81fc53369f2262cf6e85bf7db1fa4
1f1d5bfba25e3ed9cfc26bf8ab0edba9

Workflow SHA-256: af46be98e2ed11136d637310c2336ce52cf0ca3067c9ec6d11c3112038500f8e

Approved Hub scope: coordinator records, eight setup answers, own setup and own canonical model plan. Other roles' private designs, complete answers and decision history are excluded.

Purpose and shared context

Illustrative project: neighborhood shade and summer surface temperature. This is a synthetic software demonstration, not a live study or an approved research finding. Compare matched observation units, retain original records and document uncertainty before drawing conclusions.

Team and responsibility map

Kit	Role	Human operator
hub	HUB	Synthetic PI
ingestor	INGESTOR	Synthetic intake operator
spoke-a	SPOKE	Synthetic researcher A
spoke-b	SPOKE	Synthetic researcher B

Research questions and assignments

Q1

How does canopy cover relate to matched surface-temperature observations?

Lead: spoke-a | Support: spoke-b

Source: Illustrative project brief, research question Q1.

Recorded project decisions

Project Charter

Purpose

Estimate how observed surface temperature varies with canopy cover within a defined study area.

Deliverables

A documented source inventory, reproducible analysis, uncertainty summary and reviewed evidence package.

Deadlines

Six-week demonstration schedule: source review in week 1, design review in week 2, analysis in weeks 3-4, challenge review in week 5 and final package in week 6.

Constraints

Use licensed, authorized data. Surface temperature does not directly establish personal heat exposure. This demonstration contains no actual results.

Completion conditions

Complete the source audit, analysis checks and separate human review gates; retain the exact methods, artifacts and limits.

Research Logic

Questions

Q1: How does canopy cover relate to matched surface-temperature observations?

Research modes

Observational comparison with sensitivity analysis; causal claims require additional identification assumptions.

Alternatives

Season, acquisition time, land cover, spatial scale and measurement error can explain apparent differences.

Discriminating evidence

Check comparable acquisition conditions, variable definitions and predeclared alternative specifications.

Human Authority And Roles

Gate reviewers

Coordinator review: the project authority assigns workstream reviewers and documents overlap.

Independence

Record reviewer overlap instead of claiming independent review when the same person holds multiple roles.

Scope change authority

The named project authority decides changes to scope, assignments and deadlines.

Team Expertise And Capacity

Expertise

Record the relevant source, methods and review experience of each operator.

Human capacity

Budget four hours per week per example contributor; this is an illustrative assumption to confirm.

Runtime capacity

Use bounded source retrieval and the role-specific model plan; do not assume unlimited access.

Bottlenecks

Lead review and Ingestor reproduction may limit throughput. Keep tasks bounded and disclose waiting work.

Consequentiality Map

Worked examples

Publishing an unsupported causal interpretation or releasing restricted source data would require a stop and human review.

Escalation rules

Escalate material data defects, incompatible methods, access uncertainty and scope changes before affected work continues.

Method And Verification Plan

Selection rationale

Use STATISTICAL_ANALYSIS to define the estimand, sampling unit, outcome and uncertainty before selecting an analysis. Spatial dependence needs an appropriate workstream-specific review.

Verification

Audit missing values, units and joins; preserve original data; compare the primary result with documented sensitivity checks. A peer reviewer checks reasoning and reproducibility.

Ethics and control documents

Use only authorized records. Check licensing and disclosure limits. No personal identifiers are required for this illustrative design.

Provenance Policy

Depth

Retain original source identifiers, versions, retrieval dates, processing steps and parent artifacts.

Source rules

Use source documentation to establish units, temporal coverage and allowed use. Preserve conflicting or missing metadata.

Failure memory

Record failed checks, corrections and rejected assumptions with their evidence; do not silently replace prior history.

Handoff requirements

Send exact artifact versions, methods, limitations, human review and supplied conversation evidence through the approved route.

Model Recommendations

Historical design prose. The role-specific MODEL_PLAN.json is canonical.

Task strengths

Use the canonical MODEL_PLAN.json for actual available profiles and acceptance criteria.

Recommended

The structured role-specific plan supersedes this historical narrative.

Minimum

Verify adequacy for the bounded task before choosing a smaller model.

High assurance

Escalate only under the recorded task triggers and human-selected access.

Evidence and dates

This example supplies no current provider ranking or measured cost savings.

Recheck owner and trigger

The lead rechecks availability and task fit when access or requirements change.

Automation And Infrastructure Plan

Storage

Selected arrangement: an institutional shared repository with recipient-specific folders. This example does not establish any live account access.

Runtime adapter

Local Python tools and the operator-selected AI workspace; confirm actual file access before operational use.

Checkpoint cadence

Save after completed review milestones and before handoff or interruption. Each shared kit has one active checkpoint writer.

Transport

Human-mediated upload with exact package hashes and receiving-operator acknowledgment. A local file is not evidence of remote delivery.

Approved services

Repository storage is planned. Automatic messaging is not selected. The Hub is the sole publisher of approved Library additions and corrections.

Architecture Decision Record

Proposal

One Hub coordinates two Spokes and one Ingestor. Human approval remains separate from file delivery.

Risks and alternatives

Source incompatibility, review backlog and late onboarding may slow the study. Manual transfer remains available.

Unresolved decisions

Account invitations and receiving-operator access tests remain pending; the design can be reviewed before those actions complete.

Hub setup and pending actions

Automation - NOT_USED

No automatic messaging service selected.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-automation

Evidence reference: NOT_VERIFIED

Folders - PENDING

Use the generated role folder; keep private work separate.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-folders

Evidence reference: NOT_VERIFIED

Library - PENDING

Use approved Library records; publication and corrections belong to the Hub.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-storage

Evidence reference: NOT_VERIFIED

Permissions - PENDING

Account invitations and receiving access tests remain pending.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-permissions

Evidence reference: NOT_VERIFIED

Storage - CONFIGURED

Institutional repository folder selected; local example only.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-storage

Evidence reference: synthetic-local-access-receipt

Transport - PENDING

Use exact recipient-specific packages and confirm receipt.

Owner: Synthetic PI

Next action: Confirm the scoped arrangement with the named owner before operational use.

Source reference: fixture-response-transport

Evidence reference: NOT_VERIFIED

Hub model recommendation (canonical supplied record)

Acceptance check: Every reported claim must retain the supplied source reference; preserve missing values.

Comparison reference: None

Confirmation / Date: 2026-09-28

Confirmation / Human: Synthetic test operator

Confirmation / Response reference: Invented fixture response; no actual human approval

Confirmation / Status: CONFIRMED_BY_CONFIGURATOR

Cost tradeoff: Costs are unknown in this fixture; no measured savings asserted.

Default profile: base

Evidence date: 2026-09-28

Evidence reference: Invented software fixture; not a real model recommendation

Evidence status: PROVISIONAL

Lean profile: lean

No lean reason: None

No stronger reason: None

Profiles / Base / Access date: 2026-09-28

Profiles / Base / Access reference: Invented test fixture; no actual model access asserted

Profiles / Base / Host: Synthetic local test host

Profiles / Base / Model: fixture-base

Profiles / Base / Provider: Synthetic provider for software validation only

Profiles / Base / Reasoning setting: medium

Profiles / Lean / Access date: 2026-09-28

Profiles / Lean / Access reference: Invented test fixture; no actual model access asserted

Profiles / Lean / Host: Synthetic local test host

Profiles / Lean / Model: fixture-lean

Profiles / Lean / Provider: Synthetic provider for software validation only

Profiles / Lean / Reasoning setting: low

Profiles / Strong / Access date: 2026-09-28

Profiles / Strong / Access reference: Invented test fixture; no actual model access asserted

Profiles / Strong / Host: Synthetic local test host

Profiles / Strong / Model: fixture-strong

Profiles / Strong / Provider: Synthetic provider for software validation only

Profiles / Strong / Reasoning setting: high

Rationale: Synthetic fixture recommendation for hub; replace with actual task-specific evidence.

Recheck when: New task type, changed model access, repeated failure, or provider update.

Retry limit: 1

Step down when: Consider the lean profile for formatting already checked records, with output comparison.

Step up when: Ask the human to reassess after failed source traceability or unresolved synthesis contradictions.

Stronger profile: strong

Task: Coordinate the synthetic pilot; no actual scientific approval.

Intake coverage and remaining questions

Question recorded means an ask event exists. Response recorded means a current answer remains in the checklist. These checks expose omissions; they do not prove that a human read or understood the plan.

Checklist protocol: ra-architect-workflow/2

Stage	Required question	Asked	Response
1	Would you like to ask anything about the system or setup process before we begin?	Yes	Recorded
1	Which AI workspace will conduct this setup?	Yes	Recorded
1	Which verified Python 3.11-or-newer invocation will generate the kits?	Yes	Recorded
2	What project brief should this team use?	Yes	Recorded
2	What project name and stable project ID do you want?	Yes	Recorded
2	Who is the accountable project authority?	Yes	Recorded
2	Do you want person-based or role-based kit names?	Yes	Recorded
3	Do you confirm the displayed research questions with their source references?	Yes	Recorded
3	Which competing explanations must the research consider?	Yes	Recorded
3	What outputs will establish that the project is complete?	Yes	Recorded
4	Who will operate each proposed kit?	Yes	Recorded
4	What relevant expertise does each proposed operator bring?	Yes	Recorded
4	How many hours per week can each operator commit?	Yes	Recorded
4	How much review work can each reviewer realistically accept?	Yes	Recorded
4	What is the project due date?	Yes	Recorded
4	What intermediate milestones must the plan meet?	Yes	Recorded
5	Which evidence sources are available for the proposed work?	Yes	Recorded
5	What evidence could challenge the proposed explanations?	Yes	Recorded
5	Which ethical,access or regulatory constraints apply?	Yes	Recorded
6	Do you confirm the proposed research-mode mapping for each question?	Yes	Recorded
7	Do you confirm the displayed per-workstream methods and their reasons?	Yes	Recorded
7	What verification will each workstream require?	Yes	Recorded
8	Do you confirm the displayed lead/support assignments?	Yes	Recorded
8	Do you confirm the displayed workstream dependencies?	Yes	Recorded
8	Who is the single checkpoint writer for each shared kit?	Yes	Recorded

Stage	Required question	Asked	Response
9	Do the displayed workstream examples capture the consequential decisions requiring review?	Yes	Recorded
9	Who performs Gate-1 review for each workstream?	Yes	Recorded
9	Who performs Gate-2 review for each workstream?	Yes	Recorded
9	What reviewer overlaps or independence limits must be recorded?	Yes	Recorded
10	Do you confirm the proposed evidence-provenance requirements?	Yes	Recorded
10	What checkpoint cadence should the team use?	Yes	Recorded
10	Do you confirm the proposed handoff and transcript requirements?	Yes	Recorded
11	What model profiles are actually available to each operator?	Yes	Recorded
11	Do you accept the dated per-role model recommendations and their stated tradeoffs?	Yes	Recorded
11	Who will recheck model recommendations when the stated triggers occur?	Yes	Recorded
12	Which storage location will the team use?	Yes	Recorded
12	Which actual transport route will move files between people?	Yes	Recorded
12	Which required permissions have actually been configured?	Yes	Recorded
12	Which optional services do you want configured?	Yes	Recorded
12	Where should the new project folders be created?	Yes	Recorded
13	What corrections do you want to the complete architecture proposal?	Yes	Missing
13	Are any consequential decisions still unresolved in the displayed proposal?	Yes	Missing
14	Do you approve this exact displayed configuration for kit preparation?	Yes	Missing

Original setup responses (Hub scope)

Storage

Use institutional repository folders with separate recipient access. Library publication remains with the Hub.

Recorded human: Synthetic PI | Reference: fixture-response-storage

Transport

Use human-mediated uploads, exact hashes and actual receiving acknowledgment.

Recorded human: Synthetic PI | Reference: fixture-response-transport

Permissions

Hub repository access is not independently verified. Account invitations for participants remain pending.

Recorded human: Synthetic PI | Reference: fixture-response-permissions

Automation

Do not configure automatic messaging for this demonstration.

Recorded human: Synthetic PI | Reference: fixture-response-automation

Folders

Use one generated role folder per kit, with private-work, inbox, outbox and receipts.

Recorded human: Synthetic PI | Reference: fixture-response-folders

Writer

Assign one active checkpoint writer to each shared kit.

Recorded human: Synthetic PI | Reference: fixture-response-writer

Checkpoint

Save before transfer or interruption and after completed review milestones.

Recorded human: Synthetic PI | Reference: fixture-response-checkpoint

Handoff

Preserve methods, source versions, limitations, artifacts and supplied conversation evidence.

Recorded human: Synthetic PI | Reference: fixture-response-handoff

How to use this plan

This document is generated from the recorded configuration. It does not grant access or research approval. Current checkpoint records retain later progress and closed work. Revise the configuration and regenerate this view; do not maintain a separately edited PDF.

Each operator also receives their own SETUP_HANDOFF.md with setup owners and pending actions. Actual setup completion, operator acknowledgment and subsequent corrections belong in current checkpoint records. The Hub alone publishes approved Library additions and corrections.